Notes for the Poster Presentation
Topic: Customer Segmentation and A Real-World Case Study on Amazon Prime

1. Defined the Problem Statement

We started with a real-world business case: How can Amazon Prime improve customer retention and profitability by segmenting its users based on behaviour and demographics?

We framed the goal as:

Identify customer groups to improve personalization, retention strategies, and lifetime value (LTV).

2. Collected and Pre-processed Data

We worked with simulated data reflecting Amazon Prime user attributes:

- **Behavioural Data**: Streaming history, device usage, purchase frequency
- **Demographic Data**: Age, location, estimated income.
- **Transactional Data**: Subscription tenure, spend history.

We cleaned the data, handled missing values, and prepared it for analysis using Pandas.

3. Applied RFM Analysis (Recency, Frequency, Monetary)

We implemented the RFM model to score and classify users:

- **Recency**: Days since last engagement.
- **Frequency**: Number of active sessions.
- **Monetary**: Average spends on Amazon.

Each customer was scored and grouped into segments like:

- **Champions**, **Loyal**, **At-Risk**, **Potential**.

We visualized this using **Matplotlib** to show distribution and contribution.

What is RFM?

RFM stands for **Recency, Frequency, and Monetary** — it's a **customer segmentation technique** used in marketing and data analytics to evaluate and rank customers based on their behaviour.

It's especially useful for businesses with repeat customers (like Amazon Prime, e-commerce, etc.).

🔹 1. Recency (R)

- **What it means:** How recently a customer interacted with the business (e.g., last purchase or streaming activity).
- **Why it matters:** Customers who engaged recently are more likely to engage again.

Example:
User A watched Prime Video yesterday — high recency.
User B hasn't watched anything in 3 months — low recency.

🔹 2. Frequency (F)

- **What it means:** How often a customer interacts or makes a purchase.
- **Why it matters:** Frequent users are typically more loyal.

Example:
User A watches 10 shows/month — high frequency.
User B watches once every 2 months — low frequency.

🔹 3. Monetary (M)

- **What it means:** How much money a customer spends on the platform over a period.
- **Why it matters:** High-spending users contribute more to revenue.

Example:
User A spends ₹5,000/year on Amazon — high monetary value.
User B spends ₹500/year — low monetary value.

4. Implemented K-Means Clustering

We then used **K-Means clustering** (machine learning) to uncover hidden patterns in customer behaviour:

- Normalized the data.
- Found optimal number of clusters using the **elbow method**.
- Grouped users into **5 clusters** based on engagement patterns.

This helped us compare ML-driven segmentation with the RFM-based one.

What is K-Means Clustering?

K-Means Clustering is an **unsupervised machine learning algorithm** used to **group similar data points into clusters** based on patterns in the data — **without** needing labelled data.

💡 Why Use K-Means for Customer Segmentation?

It helps uncover **natural groupings** of customers based on behaviours like:

- Spend,
- Usage frequency,
- Recency of activity,
- Engagement levels, etc.

Instead of setting rules manually (like in RFM), **K-Means uses math** to find patterns and group customers automatically.

In Your Case:

You applied K-Means on customer features like:

- Recency,
- Frequency,
- Spend,
- Engagement metrics.

It grouped customers into **5 distinct segments**, such as:

- **Loyal High-Spenders**
- **At-Risk Customers**
- **Sleeping Giants**
- **New Signups**
- **Potential Champions**

These segments are **data-driven**, not rule-based — giving you deeper, more flexible insights.

Why It's Powerful

- Works great with large datasets.
- Finds hidden patterns humans might miss.
- Helps **target different strategies** for each segment.

5. Analysed Segments and Generated Insights

We interpreted both models and compared:

- **RFM Model**: Simple rule-based insights.
- **K-Means**: Data-driven, pattern-based segmentation.

We created a detailed insight table to show:

- Size of each segment,
- Revenue impact,
- Business actions (e.g., upsell, reactivate, retain).

6. Presented Business Implications

We concluded with **data-backed strategies** for Amazon Prime:

- Upsell to Champions & Potential Champions.
- Re-engage Sleeping Giants with personalized content.
- Win back At-Risk customers using campaigns.
- Target new users to boost onboarding success.

7. Tools and Technologies Used

We used:

- **Python** (for scripting and modeling),
- **Pandas** (data manipulation),
- **Matplotlib** (visualization),
- **Scikit-learn** (for clustering).

Conclusion

We combined **traditional RFM analysis** with **machine learning (K-Means)** to perform effective customer segmentation.
This dual approach gave us better clarity and actionable strategies to improve retention and revenue for a subscription business like Amazon Prime.